

TYRMA, Russian Federation

WMO: 315380

Lat: 50.0456N

Lon: 132.1770E

Elev: 1139

StdP: 14.10

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.6	-30.6	-39.4	0.6	-34.0	-35.5	0.8	-30.1	9.7	-5.7	8.2	-5.5	0.8	30	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	84.4	68.4	81.1	66.5	78.3	65.0	71.0	79.1	69.3	77.2	67.8	75.3	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.6	109.5	74.9	66.7	102.5	73.0	65.2	97.3	71.4	35.8	79.2	34.3	77.2	32.9	75.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.3	10.4	8.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-38.8	90.8	3.5	3.1	-41.3	93.1	-43.3	94.9	-45.3	96.6	-47.8	98.8		
			-38.9	74.0	3.4	2.4	-41.4	75.8	-43.4	77.2	-45.3	78.6	-47.8	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.9	-13.6	-2.1	17.4	36.3	51.9	62.3	68.0	64.5	51.8	34.4	9.3	-11.1	(6)	
(7)		DBStd	29.95	8.35	9.36	10.96	8.51	6.98	6.25	4.63	5.22	7.24	8.47	12.61	9.77	(7)	
(8)		HDD50	8583	1971	1458	1010	419	60	1	0	1	63	487	1220	1893	(8)	
(9)		HDD65	12653	2436	1878	1475	860	408	124	24	72	398	950	1670	2358	(9)	
(10)		CDD50	1625	0	0	0	9	119	369	557	450	118	3	0	0	(10)	
(11)		CDD65	218	0	0	0	0	2	43	116	55	2	0	0	0	(11)	
(12)		CDH74	1848	0	0	0	9	112	504	809	388	26	0	0	0	(12)	
(13)		CDH80	415	0	0	0	0	20	136	204	53	2	0	0	0	(13)	
(14)	Wind	WSAvg	3.3	2.1	2.8	3.7	4.5	4.5	3.3	3.1	2.7	3.1	3.9	3.3	2.3	(14)	
(15)	Precipitation	PrecAvg	25.9	0.3	0.3	0.5	1.3	2.8	3.7	5.2	5.2	3.5	1.8	0.7	0.5	(15)	
(16)		PrecMax	33.2	0.9	1.2	1.2	2.8	5.6	7.4	8.8	8.1	6.4	4.4	1.7	1.3	(16)	
(17)		PrecMin	19.0	0.0	0.0	0.0	0.2	0.9	0.9	0.5	1.9	0.7	0.6	0.1	0.1	(17)	
(18)		PrecStd	3.8	0.2	0.3	0.4	0.7	1.3	1.8	1.9	1.8	1.5	0.9	0.5	0.3	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	26.9	48.5	73.4	82.4	89.0	91.4	84.2	77.4	64.6	41.1	18.4	(19)	
MCWB			12.9	21.8	37.1	51.3	58.7	67.3	69.3	70.3	63.2	50.1	34.0	16.4	(20)		
2%		DB	7.4	22.9	42.5	63.5	76.7	83.7	85.1	81.3	72.8	56.2	35.6	13.2	(21)		
		MCWB	6.1	19.2	32.9	46.7	55.2	66.3	69.2	68.7	60.5	44.7	31.8	12.0	(22)		
5%		DB	2.9	18.9	38.3	56.5	71.9	79.7	81.6	78.3	68.5	50.8	32.0	9.0	(23)		
		MCWB	1.7	16.1	30.9	42.8	53.0	63.9	68.2	66.7	58.0	41.9	28.9	7.8	(24)		
10%		DB	-0.8	14.2	34.4	51.6	67.6	75.8	78.4	75.5	64.5	47.3	28.5	3.6	(25)		
		MCWB	-1.8	11.9	28.6	39.9	51.6	62.1	67.0	65.1	56.0	40.3	26.0	2.5	(26)		
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	22.7	37.3	53.4	60.6	71.3	73.3	73.7	65.5	51.0	35.6	16.8	(27)
MCDWB				14.8	26.4	46.4	68.6	78.4	82.4	82.8	80.9	73.5	61.7	38.7	18.3	(28)	
2%	WB		6.0	19.6	34.1	48.2	57.1	67.4	71.3	71.0	62.5	46.8	32.1	12.3	(29)		
	MCDWB		7.1	23.0	40.8	62.0	71.7	79.6	79.7	77.7	69.5	53.3	35.0	13.1	(30)		
5%	WB		1.8	16.1	31.4	44.0	55.1	65.6	69.9	69.0	60.1	43.5	29.4	7.9	(31)		
	MCDWB		2.8	18.8	37.6	55.4	67.8	76.3	78.1	75.4	66.2	49.2	32.1	9.0	(32)		
10%	WB		-1.7	11.9	28.9	40.6	53.3	63.8	68.4	67.1	57.5	40.7	25.9	2.6	(33)		
	MCDWB		-0.8	14.1	34.3	50.6	64.4	73.4	76.4	72.7	63.2	46.4	28.7	3.5	(34)		
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.5	24.6	22.7	20.6	23.3	21.1	18.5	19.7	21.6	19.1	18.6	17.8	(35)	
MCDBR			20.4	24.3	23.4	31.1	33.8	29.3	24.8	24.8	28.6	26.2	18.1	16.1	(36)		
MCWBR			19.0	21.1	15.4	16.9	15.0	12.2	10.6	12.1	15.4	15.7	14.0	15.2	(37)		
5% WB		MCDBR	20.3	23.9	21.3	28.9	28.1	25.5	20.1	19.4	22.4	22.9	17.1	16.2	(38)		
		MCWBR	18.9	20.8	14.3	16.0	13.1	11.3	9.8	10.6	12.8	14.6	13.6	15.3	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.268	0.302	0.372	0.500	0.557	0.502	0.484	0.422	0.389	0.399	0.332	0.274	(40)		
taud		2.160	2.058	1.919	1.777	1.767	2.001	2.109	2.284	2.288	2.075	2.088	2.117	(41)			
Ebn at Noon		241	259	256	234	227	243	246	257	253	218	205	215	(42)			
Edh at Noon		24	35	49	63	67	54	47	38	34	35	26	22	(43)			
(44)	All-Sky Solar Radiation	RadAvg	544	923	1398	1538	1582	1733	1599	1441	1141	749	509	422	(44)		
(45)		RadStd	14	22	77	160	153	226	138	120	117	57	40	20	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page